

4	(a) torque of a couple = <u>one</u> of the forces / a force $\times$ distance multiplied by the <u>perpendicular distance between the forces</u>	M1 A1	[2]
	(b) (i) weight at P (vertically) down normal reaction OR contact force at (point of contact with the pin) P (vertically) up	B1 B1	[2]
	(ii) torque = 35×0.25 (or 25) $\times 2$ = 18 (17.5) N m	C1 A1	[2]
	(iii) the two 35 N forces are equal and opposite and the weight and the upward / contact / reaction force are equal and opposite	B1	[1]
	(iv) not in equilibrium as the (resultant) torque is not zero	B1	[1]